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MATTERIX: toward a digital twin for robotics-assisted chemistry laboratory automation

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Accelerated materials discovery is critical for addressing global challenges. However, developing new laboratory workflows relies heavily on real-world experimental trials, and this can hinder scalability because of the need for numerous physical make-and-test iterations. Here we present MATTERIX, a multiscale, graphics processing unit-accelerated robotic simulation framework designed to create high-fidelity digital twins of chemistry laboratories, thus accelerating workflow development. This multiscale digital twin simulates robotic physical manipulation, powder and liquid dynamics, device functionalities, heat transfer and basic chemical reaction kinetics. This is enabled by integrating realistic physics simulation and photorealistic rendering with a modular graphics processing unit-accelerated semantics engine, which models logical states and continuous behaviors to simulate chemistry workflows across different levels of abstraction. MATTERIX streamlines the creation of digital twin environments through open-source asset libraries and interfaces, while enabling flexible workflow design via hierarchical plan definition and a modular skill library that incorporates learning-based methods. Our approach demonstrates sim-to-real transfer in robotic chemistry setups, reducing reliance on costly real-world experiments and enabling the testing of hypothetical automated workflows in silico.

The accelerated discovery of new materials is fundamental to addressing global challenges, such as resilient and flexible pharmaceutical manufacturing, and achieving net zero targets[1]. The urgency of these challenges demands a paradigm shift in the way that we approach materials discovery. However, materials research remains heavily reliant on manual design and execution of experimental workflows. This is slow and costly because of the inherent challenges of automating complex experimental workflows with a high degree of variability. Recent advances in laboratory automation and self-driving labs have led to the deployment of a diverse array of robotic systems to accelerate the discovery of new materials[2]. These systems have been instrumental in automating workflows ranging from solubility screening[3] and synthesis[4] to electrochemistry[5], catalysis[6], solid-state chemistry[7], chirality detection in nanocrystals[8] and the exploration of polymer electronics[9].

Most, if not all, autonomous material discovery experiments today are designed, developed and deployed directly in real-world

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environments. This approach is complex and time-consuming, encompassing algorithmic development, hardware interfacing and software integration, making it a primary bottleneck in developing fully autonomous self-driving laboratories[10]. For example, it took us 3 years to build the follow-up to our original 'mobile robotic chemist'[6], which was an autonomous robotic workflow for solid-state chemistry[7]. This was done entirely in the laboratory, without any simulations or digital twins. This approach both limits testing efficiency and impedes the progress of learning-based methods for robotic chemists, necessitating large amounts of data that are both costly to collect from real-world setups. There are inherent hurdles in integrating dexterous robots, automated platforms and multirobot system integration within the dynamic of an unpredictable mixed human–robot chemistry laboratory. As a result, the majority of robotic systems implemented in chemistry laboratories today are limited to relatively simple experimental workflows. The effective handling and interpretation of multimodal data generated during these experiments poses another considerable challenge[2]. To address these challenges, simulators offer a critical solution to these challenges by providing a safe and efficient virtual environment for development and testing. These platforms enable rapid iteration of robotic planning and control algorithms, sensor integration and experimental protocols without the constraints and risks associated with real laboratory setups. By generating synthetic data, simulators can accelerate the training of learning-based models, reducing the reliance on costly and time-consuming real-world experiments. By enabling granular, hardware-aware workflow descriptions that specify robotic actions and environmental conditions with high precision, simulators enhance experimental reproducibility, complementing widely adopted hardware-agnostic standards such as χDL (Chemical Description Language)[11].

Digital twin technologies[12] have found successful applications across domains such as manufacturing and autonomous systems, enabling process optimization, predictive maintenance and synthetic data generation through real-time monitoring and virtual–physical synchronization[13,14]. In self-driving cars, for instance, digital twins enhance safety validation and decision-making by simulating complex scenarios[15]. Beyond industrial and automotive contexts, they also show transformative potential in accelerating scientific discovery and personalized medicine[16]. Despite these advances, the adaptation of digital twins to self-driving laboratories—where they could address critical challenges in lab automation, reproducibility and experimental design—remains underexplored.

For this purpose, we propose MATTERIX. This is a multiscale robotic simulation framework (Fig. 1) for developing digital twins of laboratories, to facilitate the design and evaluation of experimental workflows and to allow easy deployment to the real environment. MATTERIX is a comprehensive set of open-source tools building on the Nvidia Isaac Sim physics engine[17] and Isaac Lab framework[18]. Because it is based on Isaac Sim and Isaac Lab, MATTERIX comes with inherently fast and accurate physics simulation, parallelized graphics processing unit (GPU) simulation, contact-rich interactions (such as signed distance field-based collision meshes for precise collision detection[19] and convex decomposition for simplified collision shapes[20]), application programming interfaces (APIs) for object definition and a range of robot learning libraries (Stable Baselines 3 (ref. 21), robomimic[22], RL-Games[23], RSL RL[24] and SKRL[25] among others). We further extend the framework with several features, including a semantics engine, that is, a system enabling the modular simulation of behaviors, such as chemical reaction kinetics, heat transfer and logical processes (which are not inherently supported by robotics simulators). Through these features, MATTERIX addresses the current limitations of the suite of tools available for chemistry laboratory automation as showcased in Fig. 2, which highlights the requirements from a digital twin of a chemistry laboratory and compares MATTERIX with the recent works in this domain. The advantage of using MATTERIX is it adds many of the key features required for chemistry laboratory automation, including particle simulation, simulation-to-reality deployment in real laboratories and workflow verification, which is imperative in safety-critical environments, as is our domain application within a single framework. Simulation environments such as those detailed in refs. 26–30 showcase the role that physics-based simulations can play for accurately modeling environments used for robotic chemist skill learning and developing better perception systems. Other simulators are more tailored for developing and validating automated synthesis workflows or automating research operations, and recent works[31,32] underscore their role in streamlining material discovery.

MATTERIX aims to be a comprehensive tool that embeds key features for automating chemistry experiments toward empowering automation chemists to embrace digital tools for accelerating the design and deployment of automated experiments. It is also aimed to be a tool for collecting extensive datasets required for advancing robotic manipulation, refining control algorithms and optimizing experimental workflows, all within a highly controllable and efficient setting.

In particular, our contributions are the following:

(1) Multiscale and GPU-based chemistry experiment workflow simulation. This is an open-source simulator that can simulate chemistry laboratory experiments. MATTERIX extends NVIDIA Isaac Lab[18], the physics engine of NVIDIA Isaac Sim and its particle-based simulation with a semantics engine. The multiscale nature of MATTERIX allows the design and verification of full chemical experiment workflows.
(2) Easy generation of new chemistry laboratory environments. MATTERIX allows users to easily design new environments by identifying the set of assets and their locations (poses) in the digital twin environment. It comes with an extensive library of digital and photorealistic assets. These assets are either curated from open-source resources or custom-built using computer-aided design and three-dimensional (3D) scene reconstruction techniques. This includes diverse glassware, (mobile) robotic manipulators, automated platforms and other essential laboratory equipment, facilitating the easy setup of detailed simulation environments. Moreover, it allows users to generate digital twin environments based on real environments using 3D perception and scanning methods, interfaced with laboratory devices and robots.
(3) Design of workflows for flexible chemistry experiments. MATTERIX enables users to easily build chemistry workflows. This has been enabled using a tree-structure hierarchical state machine and a library of skills that allows users to combine those skills to control multiple automation devices and robots (agents) in the digital twin environments. These skills are based on several methods such as reinforcement and imitation learning, classical motion planners with/without obstacle avoidance and whole-body mobile robot controllers. The combination of hierarchical state-machine-based and learning-based methods to demonstrate policies for various laboratory tasks facilitates the reuse of learned skills in novel experimental workflows. The combination of computer vision and safe motion planners with obstacle avoidance allows users to transfer samples and objects flexibly without the need for hard-coded robot trajectories.
(4) Deployment of workflow solutions to real laboratory setups. We demonstrate sim-to-real transfer for autonomous chemistry laboratories, deploying general-purpose robots and automated platforms programmed using MATTERIX in a real-world laboratory setting. These experiments validate the repeatability and transferability of our approach, paving the way for broader laboratory automation.

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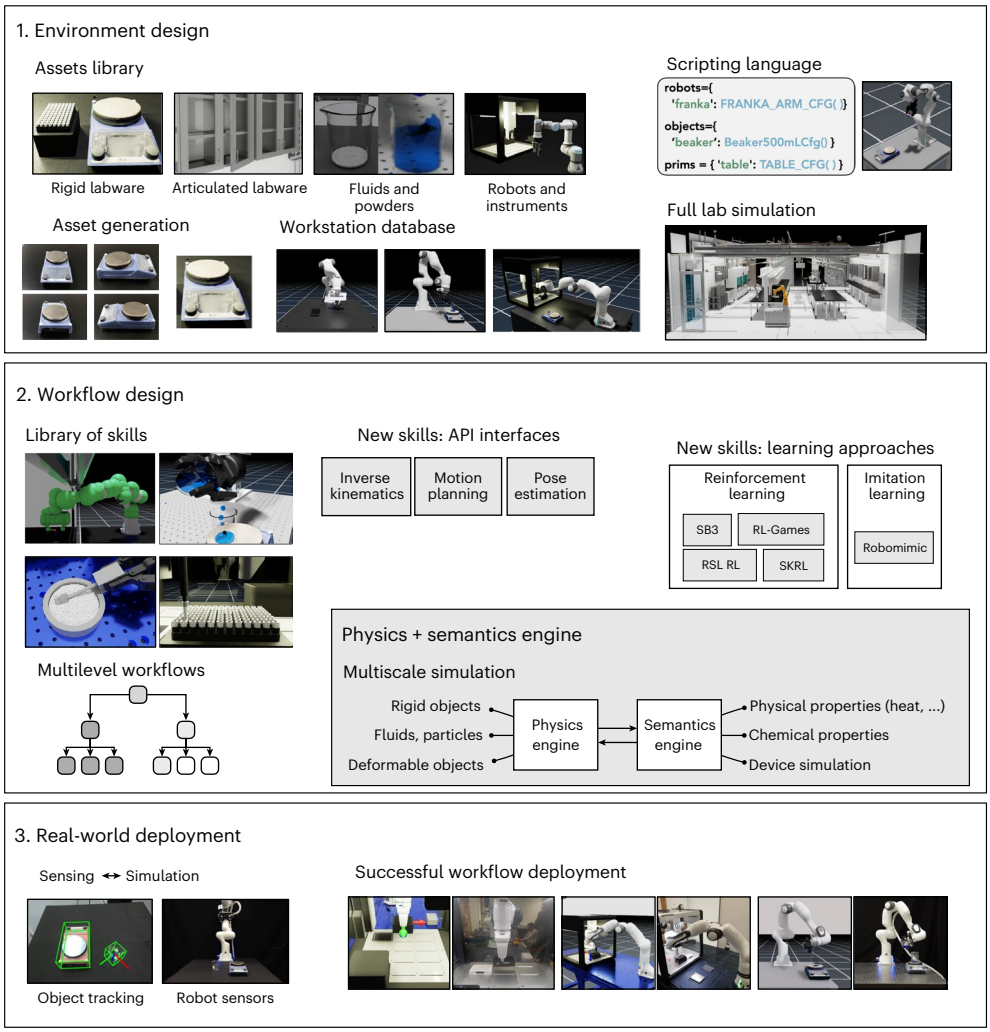


Fig. 1 | MATTERIX architecture and components. MATTERIX generates a digital twin environment by specifying objects and their poses using a library of digital and photorealistic assets. (1) It solves a given workflow to perform the chemistry experiment. (2) To perform the experiment, it trains new policies that are not available in the skill library but are required for experimenting. A multistep workflow is generated by a human or large language model. Using a database of known reactions and chemical kinetics approximation, MATTERIX verifies if the target chemical is synthesized. MATTERIX is deployed in the real world and performs a chemistry experiment. (3) The object poses are perceived using a camera stream in the real environment, and the simulated robots are matched with the real robots.

Results

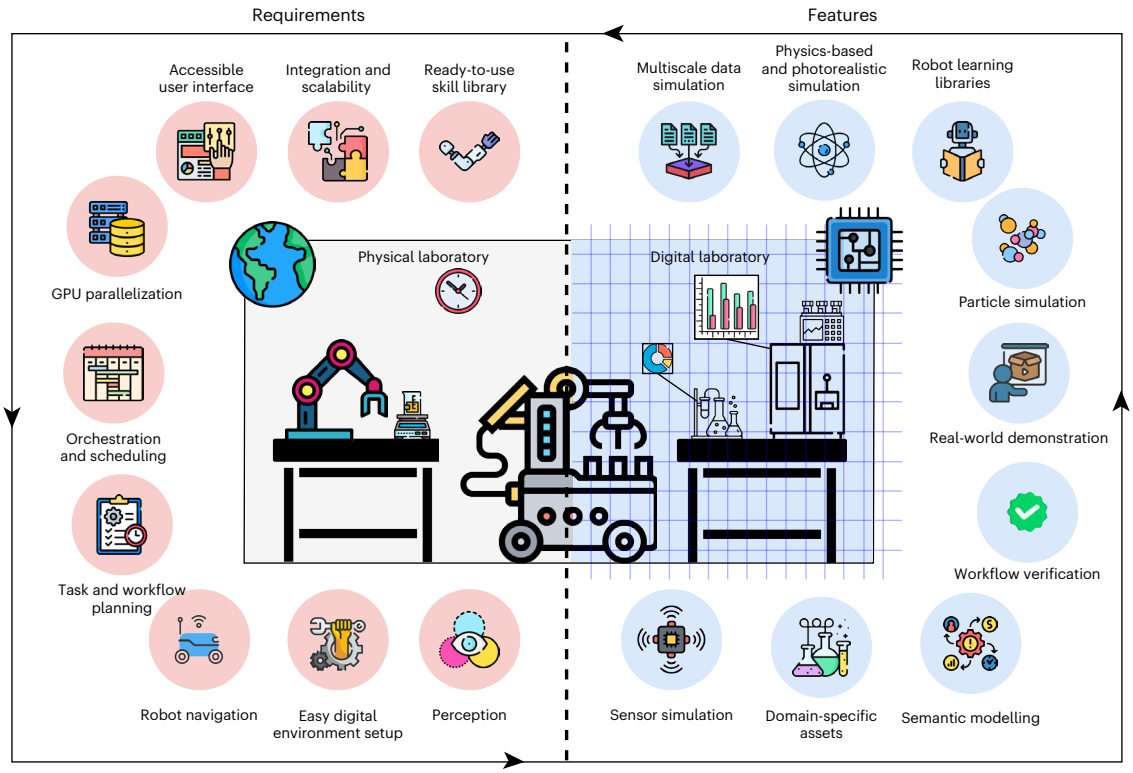
We assess the performance of different components of MATTERIX, including the generation of digital environments and a full self-driving laboratory setup, skill library and semantics engine. Following that, we demonstrate how MATTERIX simulates two different multistep chemistry experiment workflows (one organic, one inorganic). In addition, we demonstrate and evaluate the deployment of our method from simulation to a real-world setup, showcasing its ability to execute a long-horizon, contact-rich chemistry experiment with a single robot arm and to conduct multisystem experiments using a robot arm and an automated liquid handler.

Virtual environment generation results

MATTERIX provides an open-source library of 3D assets that are commonly found in chemistry laboratories. This consists of 2 pieces of laboratory equipment for contained experiments (glove box and fume-hood), 2 fixed-base robot arms (7 degrees of freedom Franka Emika Panda and 6 degrees of freedom UR3e), 1 mobile manipulator (KUKA KMR iiwa), 1 benchtop liquid handler (Opentrons, OT-2), 12 different laboratory objects (beakers, pipettes, graduated cylinders, spatulas, well reservoirs, well plates, glass bottles, flasks, funnels, racks, test tubes and vials), 2 laboratory instruments (stirring/heating plate, scale and syringe pump), 2 laboratory tables and 1 full 3D model of a laboratory environment. This list is not exhaustive, and users can extend it by adding their own assets. MATTERIX also enables the addition of new assets using neural radiance fields (NeRF), a 3D computer vision technique for scene reconstruction that allows the synthesis of photorealistic novel views[33]. Figure 1 illustrates a virtual asset of a laboratory instrument created by the NeRF model, using an object-centric video provided by the user. This approach is particularly useful for constructing a universal scene description (USD) representation of an object from a real-world setup when the corresponding asset is absent in the asset library. The physical properties of the generated asset, such as mass or friction, are manually specified by the user. Future research could explore learning methods and physical robot interactions with objects to automate or refine this process[34,35]. MATTERIX also supports the definition of multiple, heterogeneous embodied agents (robots) in an environment, facilitating their collaborative execution of complex autonomous laboratory experiments.

All these assets are modeled by USD[36]. USD is a high-performance, extensible platform for collaborative 3D scene construction, enabling robust asset interchange, composition and scalability across

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Features	MATTERIX	Chemistry 3D[26]	Powder weighing[27]	Sample scraping[28]	Stir to pour[29]	ChemGym RL[31]
Multiscale data simulation	✓	✓	×	×	×	×
Physics-based and photorealistic simulation	✓	✓	✓	×	✓	×
Robot learning libraries	✓	✓	×	×	×	✓
Particle simulation	✓	×	×	×	✓	×
Real-world demonstration	✓	×	✓	✓	✓	×
Workflow verification	✓	×	×	×	×	✓
Semantic modeling	✓	×	×	×	×	×
Domain-specific assets	✓	✓	×	×	×	×
Sensor simulation	✓	✓	×	×	×	×

Fig. 2 | Digital twin requirements and comparison with MATTERIX. Top: key requirements and benefits of digital twins in chemistry laboratory automation. Bottom: a comparison of MATTERIX with related works. Credit: icons, Flaticon.com.

animation, visual effects and digital content creation. Assets specify physical properties (mass, friction and contact parameters), textural/visual representations (color patterns, roughness and bump maps), object hierarchies, transformations (position, rotation and scale), material properties (interaction with light, shaders and reflectivity) and metadata (object names, labels and other descriptive information). Moreover, the user can attach additional properties to an asset in the asset library and query them during the simulation. For example, the user can specify a list of object key points (place to grasp, location to perform insertion and so on) and use that information for solving automation tasks.

Fluids and powders are fundamental components in chemistry laboratories—most chemicals, other than gases, tend to be either fluids or powders. MATTERIX provides users with the capability to easily incorporate fluids and granular media into their chemistry workstation designs, enabling more realistic physical simulations of their experiments. To add liquid or powder in the simulation, users can specify the quantity and location for initialization. Particle properties, such as physical behavior or appearance, can also be customized beyond the default settings. They are simulated using a GPU-accelerated position-based-dynamics particle simulation[37], permitting fast and accurate computation and accounting for their interaction with all other objects in the scene.

Workstation virtual environments. To create and initialize a laboratory workstation, the users can add the assets from the asset library to a configuration file, which MATTERIX would then use to generate a virtual environment. To achieve this, one simply needs to create a dictionary, with the asset naming and typing. Figure 3 demonstrates different chemistry laboratory workstations in MATTERIX. These environments enable users to design, develop and deploy modular robotic and chemistry workstations toward advancing the adoption of self-driving laboratories. Although simpler environments are well-suited for training robot policies, more complex environments, which incorporate a wider range of laboratory equipment and realistic layouts, are essential for conducting thorough feasibility studies, optimizing laboratory layout designs for efficiency and implementing comprehensive monitoring systems for automated experiments. A full list of environments and assets and their functionalities are provided in Supplementary Sections 1 and 2.

Figure 3 (bottom) illustrates an example of a liquid handling station and the script to initialize the environment. The environment would comprise different robots, for example, a Franka Panda arm equipped with a Robotiq 2F-85 adaptive gripper and an OpenTrons OT-2 benchtop liquid handler. Several key objects need to also be defined, including an OpenTrons pipette tip rack, a six-well wellplate, an IKA RET

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Fig. 3 | Examples of MATTERIX digital twin full chemistry laboratory environment and its workstations. A chemistry laboratory simulation framework comprising different (mobile) robotic chemists, instruments and glassware across various laboratory tasks, including materials and fluids (shown as A–F). Bottom: how an environment is created.

(a magnetic stirrer with integrated balance) control scale and a two-well reservoir. The specification of locations of numerous objects in chemistry labs can be challenging, and to address this challenge, MATTERIX provides a utility function that allows users to initialize environments with a USD with several objects. For example, in the liquid handling station, the 96-pipette tip with the rack is defined as a nested multiobject USD.

Full self-driving laboratory environment. The digital twin of a full chemistry laboratory is shown in Fig. 3. It is modeled after one of the Acceleration Consortium's inorganic self-driving laboratories at the University of Toronto. The laboratory features four fume hoods, one benchtop liquid handling station, one solid dispensing station, one glove box for air-sensitive and moisture-sensitive reactions and one washing and drying station. Robotic systems include five

fixed-base manipulators (Franka Emika Panda and Universal Robots) and one mobile manipulator (KUKA KMR iiwa). Users can define and execute long-horizon tasks and workflows within this digital twin environment.

Manipulation and perception skills

MATTERIX automates chemical experiments by providing a library of reusable and goal-conditioned skills. These skills are generated through a combination of manipulation and perception methods. A hierarchical state machine organizes these skills, with leaf nodes representing primitive actions, and higher-level nodes representing skills composed of these primitives. Similarly, workflows are constructed by hierarchically combining these skills. This layered approach enables multiple levels of abstraction, facilitating collaboration between chemists and roboticists. Roboticists can focus on developing and refining the library of abstract skills, whereas chemists can leverage these skills to design and implement their specific workflows.

Figure 4a illustrates a three-layer hierarchical state machine used to pick up an object and place it on top of another. This state machine is generic and can be reused by a chemist to perform pick-and-place operations across different workflow stages. The workflow proceeds from left to right and is considered complete when all actions in the state machine are successfully executed. In the diagram, the dark gray indicates completed actions, light gray shows ongoing actions and white denotes actions not yet executed.

Manipulation skills. MATTERIX includes various motion planning and learning methods as primitive actions, serving as the fundamental building blocks for enabling manipulation skills. These methods include: (1) a robot joint-level controller, (2) a robot motion planner with an inverse kinematics solver, (3) grasping and ungrasping, (4) a whole-body controller for mobile robots, (5) relative motion in Cartesian space, (6) a waiting primitive action, (7) the cuRobo constrained motion planner with obstacle avoidance[38] and (8) a check-point primitive action for deploying trained policies (via reinforcement learning or imitation learning) on robots or automation devices. These primitive actions can be optionally combined within a hierarchical state machine to achieve complex manipulation skills.

Figure 4 demonstrates a mobile 'robot chemist' performing various manipulation skills in conjunction with other robots. Figure 4a illustrates the robot picking up a well plate using the inverse kinematics robot motion planner and placing it inside the OT-2 liquid handler. This example showcases how different primitive methods can be combined within a hierarchical state machine to achieve the abstract manipulation skill 'Pick&Place(obj1, obj2)', which involves picking up 'obj1' and placing it in 'obj2'. When defining the workflow, a chemist can specify the actual objects. Figure 4b shows the Franka robot picking up a vial and inserting it into a vial rack hole. Figure 4c presents an instance of the cuRobo motion planner reaching for a beaker inside a fume hood. Figure 4d demonstrates the KUKA KMR iiwa mobile robot transferring samples between two workstations using a whole-body motion controller. Figure 4e–h illustrates robots performing various tasks, including pouring liquid into a beaker, weighing and scooping powders and handing over a beaker using two robotic arms.

Perception. Moving beyond traditional laboratory automation setups that rely on hard-coded instructions and the use of AprilTags or QR codes, MATTERIX integrates FoundationPose[39], a foundation model for model-based six-dimensional object pose estimation and tracking. By leveraging FoundationPose, MATTERIX reduces the need for tedious manual calibration typically required in chemistry laboratories to integrate multiple devices and robotic systems. When used as the digital twin counterpart of a real-world setup, FoundationPose allows MATTERIX to determine the object's position and orientation within the digital twin configuration. This removes the need for users to manually specify object poses within the robot's workspace (as illustrated in Fig. 4i). Furthermore, when users define an action-such as picking up an object-the estimated object pose from the perception module can be used to determine the grasping position. Given that all assets within the digital twin possess their respective USD 3D models, these models can be used as input for FoundationPose.

Integrated physics and semantics engine

In MATTERIX, we developed a semantics engine to simulate behaviors that are challenging to model or currently unsupported by Isaac Sim's physics engine. The semantics engine extends the n-dimensional physics state vector $\mathbf{x}(t) \in \mathbb{R}^n$ by introducing a set of logical semantic states $\mathbf{l}(t) \in \{0, 1\}^m$ and continuous semantic states $\mathbf{s}(t) \in \mathbb{R}^p$. Events and processes are linked to changes in both instantaneous and continuous semantic states. These changes are functions of the physics and semantic states, as well as the input action vector $\mathbf{a}(t)$ from the workflow. The input action vector includes both continuous physics input actions (robot joint efforts) and discrete semantic actions (turning on a heater and setting its target temperature). Both the continuous physics actions and the discrete semantic actions are updated directly on the GPU, enabling high-throughput control and simulation. Events can occur either as the physics and semantics engines advance the simulation (two objects coming into contact) or in response to an input action from the workflow (activating the heater). Processes represent continuous behaviors that typically require certain conditions to be met to take effect. For example, to aspirate a liquid into a liquid handler tip, the pipette tip must be inside the solution. Similarly, increasing the temperature of a heater requires that the heater be turned on. Moreover, semantic states can influence physics states. For instance, when a pipette tip is loaded, it will move in sync with the liquid handler's pipette motion. Together, the physics and semantics states are simulated using $\mathbf{y}(t+1) = \mathbf{G}(\mathbf{y}, \mathbf{a})$, where $\mathbf{y} = [\mathbf{x}, \mathbf{s}, \mathbf{l}]$ is the extended physics and semantics state vector, and $\mathbf{G}$ is the integrated physics and semantics dynamics.

MATTERIX natively supports the semantic behaviors and functionalities found commonly in chemistry laboratories. It simulates heat transfer and known first-order chemical kinetics using information provided by domain experts (chemists). Moreover, it simulates various device functionalities. The semantics engine is modular and can be easily extended by users. Figure 4j,k illustrates example behaviors and device functionalities enabled by the semantics engine. Figure 4j shows the OT-2 liquid handler's ability to load and remove tips. When the user commands the OT-2 to load a tip, the pipette link moves toward the tip. Once the pipette is inserted into the tip, the loading occurs. After dispensing, when the user commands the OT-2 to remove the tip, the pipette moves to the OT-2's disposal area and detaches the tip. Figure 4k shows the results of aspirating and dispensing solutions during the OT-2's liquid handling process. Moreover, Fig. 3f demonstrates a mobile robot turning a knob, causing water to flow from the faucet in the washing station.

The semantics engine is implemented using PyTorch, which enables GPU parallelization. Figure 4l demonstrates the computational performance of MATTERIX when the heat transfer semantics engine is added to an environment consisting of a Franka Emika robot arm, an IKA RET control heater and a beaker. The simulations ran on a system with an Intel Xeon W-2235 CPU (3.80 GHz, 12 cores), 31 Gi RAM, an NVIDIA RTX A6000 GPU and Ubuntu 20.04.6 LTS (x86_64). To compare the computational performance with and without the semantics engine, the following parameters are set randomly: the robot's movements and the IKA heater's target temperature (ranging between 40 °C and 100 °C). The results show that for 2,048 and 4,096 parallel environments, after running 1,000 simulation steps, the GPU overhead rises by only 2.97% and 3.42%, respectively, when the heat transfer semantics engine is enabled.

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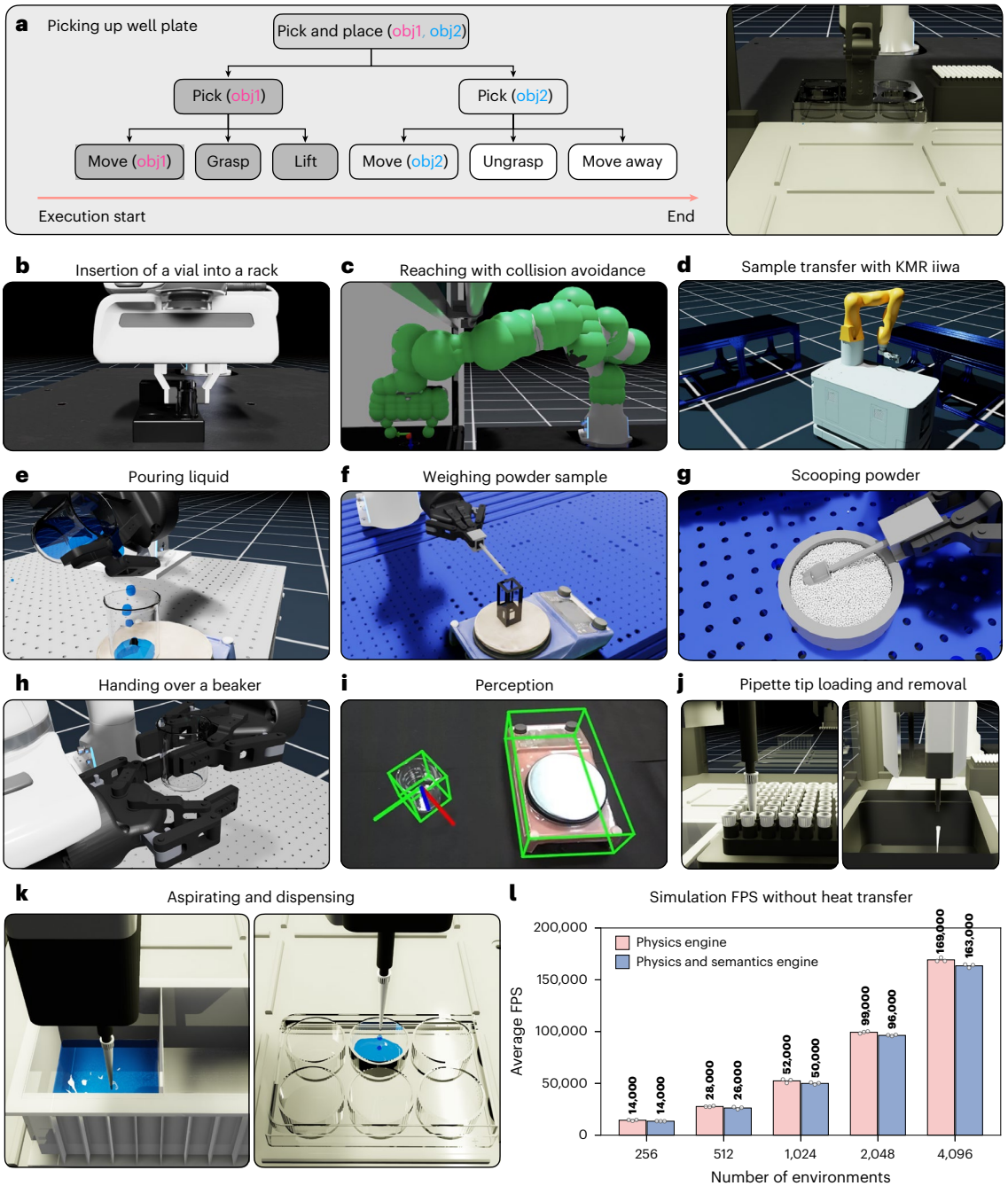


Fig. 4 | A library of manipulation, perception skills and device functionalities in MATTERIX. The figure demonstrates the behavior and performance of our integrated physics and semantics engine. **a**, The creation of abstract skills using a hierarchical state machine. **b**–**h**, Various skills for manipulating rigid and granular objects. **i**, Object pose estimation via computer vision. **j**,**k**, Liquid handling device functionalities enabled by the integrated semantics engine. **l**, Simulation throughput for a heat transfer environment with and without the semantics engine. For 2,048 and 4,096 environments, the GPU overhead with the semantics engine results in frames per second (FPS) reductions of only 2.97% and 3.42%, respectively, compared with the physics-only engine.

Simulation of multiscale chemistry workflows

One primary purpose of digital twins is to develop and evaluate environments and workflows for chemistry experiments. Figure 5 shows the results of a multiscale simulation for chemistry experiments, combining multibody dynamics and particle simulation via the physics engine with heat transfer and chemical kinetics via the semantics engine. Supplementary Section 5 outlines the experimental workflows required to execute the procedure defined by the hierarchical state machine.

Figure 5a illustrates an organic chemistry experiment where the unimolecular nucleophilic substitution reaction occurs between ethanol (EtOH) and hydroiodic acid (HI), heated on the IKA heater. The reaction only proceeds if the preconditions are met. This includes the requirement for the concentrations of the reactants to be non-zero and the temperature of the solution to be within a user-specified range (in this example, 30–70 °C). The reaction mechanics are governed by user-specified parameters, which can specify a constant k reaction rate or the parameters of a temperature-dependent $k(T)$ on the basis of the Arrhenius equation[40]. The temperature range precondition is still enforced to constrain the reaction to physically relevant temperatures. In Supplementary Fig. 3, we simulate the same process with a heater temperature of 40 °C. At this much lower temperature, the reaction

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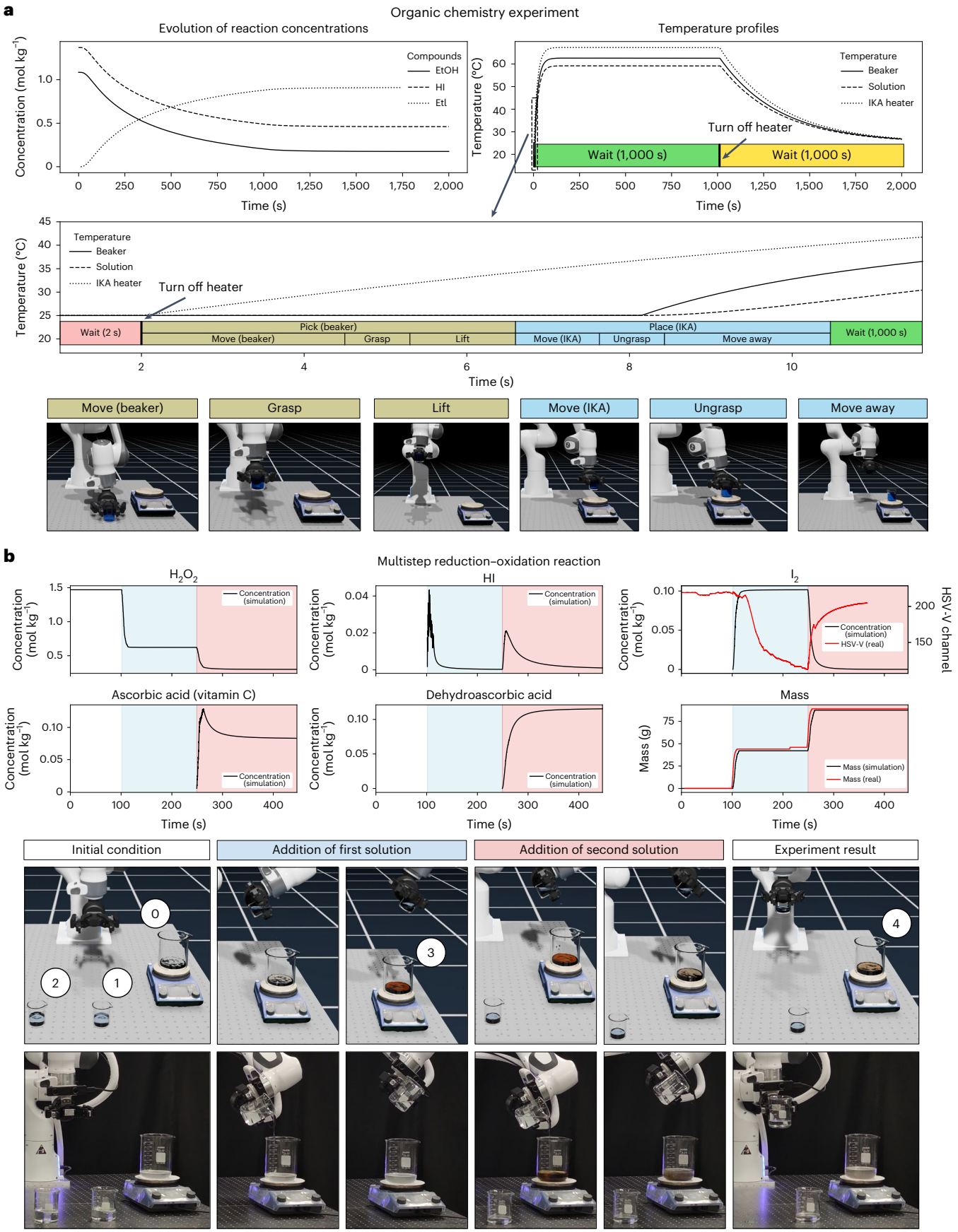


Fig. 5 | Multiscale simulation of chemistry experiments. a, A single-step organic chemistry experiment demonstrating the interaction of physical manipulation, heat transfer and chemical kinetics simulation. **b**, A two-step reduction–oxidation chemistry experiment illustrating the interaction of physical manipulation and chemical kinetics simulation. See Supplementary Section 6 for the video of the experiment in **b**. HSV-V, the value channel (V) of the HSV color space. The V channel represents pixel brightness and is used to measure color changes resulting from the reaction.

rate of the substitution reaction is slower, and less product is produced by the end of the simulation.

In the workflow, the heater turns on after a 2-s delay, causing only the heater's temperature to rise. At around 8 s, the robot places the beaker with the solution on the heater's hot plate, causing the beaker and solution temperatures to increase. Energy dissipation to ambient air affects all objects involved in the heat transfer simulation. As the solution warms, the reaction begins at 11.4 s and continues until 1,639.9 s, as long as the temperature stays within the reaction's permissible range. The heater turns off at 1,010.6 s, and the solution gradually cools due to ambient air dissipation. After 2,000 s, the experiment concludes with a solution containing EtI at a concentration of 0.91 mol kg^{-1} and a temperature of 26.65 °C.

Figure 5b illustrates a multistep redox experiment sequence, demonstrating the capabilities of the chemical kinetics simulation in handling workflows involving multiple solutions and reactions that involve both inorganic and organic transformations. The reaction occurs at ambient temperatures, and three solutions were prepared in separate beakers. The reaction starts by adding hydrogen peroxide (2.1 mmol H_2O_2 in 48 g H_2O from beaker 1) with iodide (4 mmol I^- in 100 g H_2O in beaker 0). This addition oxidizes iodide, leading to the formation of iodine (I_2), which forms dark blue–brown complex with the starch indicator over the course of 250 s. In the subsequent step, ascorbic acid (2.4 mmol ascorbic acid in 48 g H_2O from beaker 2) is added to reduce I_2 to I^-, leading to the consumption of the iodine–starch complex, and the dark blue–brown color of the reaction mixture fades away. Figure 5b, bottom, shows the real experiment alongside the simulated experiment. The real experiment results are displayed in the plots with red lines, representing mass changes measured by the IKA RET control scale. We also tracked the reaction progress using a camera stream focused on the solution; as the first reaction proceeds, the brightness in the hue–saturation–value (HSV) channel changes. As shown in the results, the first reaction occurs more quickly in the simulation than in the real experiment, highlighting the importance of online learning and parameter adaptation in the digital twin based on real experimental data.

Deployment to real laboratory environments

We evaluated MATTERIX's ability to deploy long-horizon experiments from simulation to a real-world laboratory setting. As shown in Fig. 6, we conducted three experiments: (1) Object Pick and Place: we assessed MATTERIX's robustness in object pick-and-place tasks using perception—a common manipulation skill in chemistry laboratories; (2) liquid handling: we demonstrated MATTERIX's ability to transfer robot manipulation skills for handling liquids; (3) long-horizon experiment: we tested MATTERIX in a real chemistry laboratory sample preparation setup, where the robot sets up a liquid handling station with an OT-2 automation device.

Object pick and place experiment. A key task for robots in chemistry laboratories is picking and placing various objects and laboratory consumables, particularly transparent glassware. Figure 6a shows MATTERIX performing a pick-and-place task with a beaker arbitrarily placed on a IKA plate in front of the robot as part of an experiment. When the digital twin starts, the simulated robot syncs with the real robot. The robot observes the real scene to estimate object locations using FoundationPose and updates its digital twin. As the robot moves in simulation, the real robot follows these movements. For this experiment, we used a state machine with inverse kinematics motion planner skills. Across 12 trials, the task execution achieved a success rate of 75% (9 successful attempts). Failures occurred in three cases: twice due to incorrect beaker picking and once due to misplacement of the beaker on the IKA plate. These failures stemmed from multiple factors, including inaccuracies in object pose estimation, robot action errors, errors in camera calibration, object frame definitions and mismatches between digital and real assets, such as size discrepancies. Developing robust pick-and-place algorithms and reducing the sim-to-real gap remain active research directions. Addressing both human-in-the-loop sources of error and robot-level uncertainties is critical for improving autonomous reliability in real-world deployments.

Liquid pouring experiment. Figure 6b presents the robot's execution of a fundamental laboratory task: pouring fluids into laboratory glassware. Here, the robot is pouring water into a beaker placed on an IKA RET control device. Initially, the robot grasps a water-filled beaker and repositions it for pouring. Subsequently, a controlled rotational motion is executed to dispense the fluid into a target beaker positioned on an IKA RET control device. The robot then returns to its home configuration, completing the fluid pouring task. A hierarchical state machine was used to run the experiment with an inverse kinematics controller. In total, we conducted ten experiments, with the beaker placed at arbitrary locations within the workspace. The robot estimated the beaker's position using FoundationPose. We achieved a 90% success rate in real-world experiments. One failure occurred due to a large error in the beaker pose estimation, which caused the robot to grasp the beaker incorrectly, resulting in the beaker sliding. In addition, there was a case where the simulation failed but the real experiment succeeded: the beaker fell off the table in the simulation because the simulated table size did not match the real one. This highlights the importance of accurately matching the simulated environment to the real robot workspace. The amount poured in the real experiment, measured using the IKA plate, was 59.6 ± 7.9 ml out of an initial volume of 101.77 ml, whereas in the simulation it was 61.3 ± 4.4 ml out of 100 ml. The difference in pouring amounts between the simulation and the real experiment stems from differences in the initial liquid volumes, slight variations in beaker sizes and the differing behavior of the liquid in the physical world versus the simulation.

Liquid handling station. This experiment demonstrates MATTERIX's capability in robotics-assisted liquid dispensing. Figure 6c shows the Franka Research 3 (FR3) robot setting up the station by placing the dispensing rack, tips rack and reservoir well in the OT-2 tray. The OT-2 head then attaches to the pipettes before moving to the dispense rack to start fluid dispensing.

MATTERIX programs the task in simulation, and the policy is transferred zero-shot to the OT-2 and FR3. The policy was implemented as a hierarchical state machine with hand-tuned low-level controllers and inverse kinematics, developed and refined entirely in simulation to ensure robust deployment on the real system. The real FR3 and OT-2 are actuated using panda-py[41] and ot2_driver[30], respectively.

Discussion

This work introduces a multiscale digital twin simulation framework for chemistry laboratories. Although there are alternative robotic simulation versions for chemistry laboratory tasks, most focus on a single task[27–29], and the most comprehensive one so far lacks heterogeneous robots with automated platforms and real-world sim-to-real transfer[26]. In comparison, MATTERIX distinguishes itself through its integrated semantics engine, hierarchical workflow representation and easy sim-to-real transfer of developed systems. In addition, it simplifies usability by offering an extensive library of assets, manipulation skills, and perception tools for defining digital twin environments and generating workflows for chemistry experiments.

However, the current iteration is not without limitations. The reliance on approximated chemical reaction models within the semantics engine can introduce inaccuracies. Integrating quantum chemistry computational methods and machine learning could strike a balance between accuracy and the efficiency required for real-time simulation[42]. Currently, MATTERIX does not support gas-phase simulation in chemical reactions; future versions aim to incorporate this using

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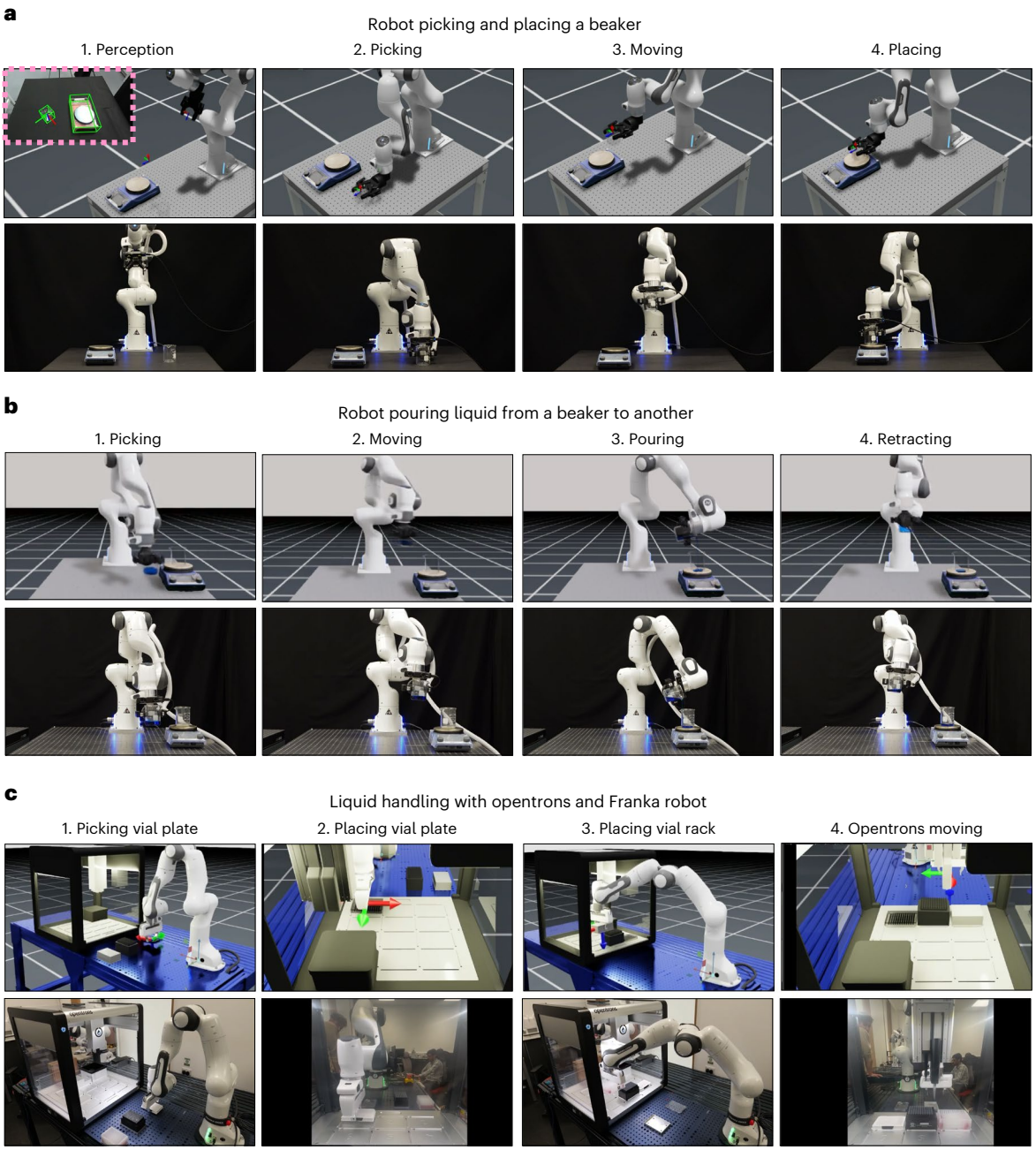


Fig. 6 | Deployment of workflows to the real setup. These experiments demonstrate the sim-to-real transfer of workflows developed in the digital twin. **a**, The robustness of pose estimation and a pick-and-place task, repeated ten times as a common laboratory automation task. The dashed pink box in the top-left corner shows the camera view used for perception, along with the estimated object bounding boxes obtained via pose estimation. **b**, The robot chemist transferring liquids during a pouring task. **c**, A real-world chemistry laboratory with the robot chemist and a liquid handler working together seamlessly. Refer to Supplementary Section 6 for videos of the experiments.

Isaac Sim's physics engine and semantics integration. Moreover, the design and computational cost of simulating long-horizon experiments remains a challenge, especially for successfully planning extended workflows. Incorporating large language and vision models could lower the barrier to entry, increasing the accessibility of MATTERIX for designing digital twin environments and automating workflow planning[43]. Regarding workflow planning, while hierarchical state machines offer an intuitive and user-friendly way to define workflows, they have limitations in deployment-particularly in recovering from failures or generating contingency plans. To address these challenges, more flexible frameworks such as behavior trees[44] or and-or graphs[45] could be adopted. Furthermore, the reliance on FoundationPose may introduce inaccuracies in object pose estimation, affecting real-world deployment precision. Refinement of object pose estimation techniques, potentially through the integration of multimodal sensor data, will improve real-to-sim deployment performance.

In addition to improving workflow flexibility and perception accuracy, addressing real-world experimental failures, such as clogging or equipment malfunction, is critical for robust and safe deployment. These failures often arise from discrepancies between simulated and physical environments, caused by differences in states, sensor noise, model inaccuracies, or limitations of the physics and semantics engines. Future versions of MATTERIX will integrate real-time feedback from IoT sensors, such as temperature probes, flow meters, and

visual monitoring, to detect deviations between simulation and reality. These measurements can trigger safety protocols, flag simulation inaccuracies, and guide model refinement[46]. By incorporating online learning methods, surrogate models trained from real-world data can be embedded into the digital twin to improve fidelity and enable predictive maintenance strategies that proactively address failure scenarios[47].

Future work could expand the skill library to include manipulation policies for soft materials, such as tubing, and delicate glassware. Moreover, large language models (LLMs) could utilize the skill library to procedurally generate and execute hierarchical state machine plans in simulation, forming a data-generation pipeline for training experimental planner policies[48–50]. Beyond this, this framework could facilitate the co-design of chemistry workstations and laboratories to optimize performance metrics such as throughput while also serving as a tool for data collection to train experimental planning models.

Beyond automation, digital twins could serve as platforms for scientific exploration and discovery. By enabling parallel simulations and "what-if" experimentation, they support the development of AI-driven experimental planners and hypothesis testing in silico. In domains where physical processes, such as fluid handling or reaction kinetics, can be modelled with reasonable accuracy, digital twins allow exploration of new protocols and experimental conditions. Despite current limitations in simulating complex chemistry or multiphase interactions, these platforms offer a foundation for training robust robotic policies that can later be deployed for real-world scientific discovery and experimentation.

We believe MATTERIX is a powerful tool for accelerating chemical discovery and foresee its application in adjacent fields, such as biotechnology. It represents a substantial effort to bridge the robotics and laboratory automation communities by offering a platform that could be used to create benchmarks for testing robotic approaches in the laboratory automation application domain. Moreover, it provides a tool for the laboratory automation community to leverage these approaches, enabling the development of flexible, robust, and safe algorithms, as well as data collection, testing and monitoring for chemistry laboratory automation and scientific discovery. Future developments will further solidify its role as a key resource for researchers and industry stakeholders, accelerating the deployment of automated robotic chemistry experiments. Toward this goal, we encourage open-source contributions to expand its capabilities and foster collaboration.

Methods

The components and architecture of MATTERIX are presented in Fig. 1. MATTERIX constructs the simulation environment using user input and an asset library, then simulates multiscale chemistry experiments in the digital twin while verifying the workflow. For real-world deployment, MATTERIX captures environmental images and robot proprioception data to execute policies across laboratory devices and robotic systems.

The simulation framework is formulated on the assumption that a set of agents, in our case robotic chemists, interact with the world-the chemistry laboratory environment-receiving observations and acting through computed controls. The world comprises robotic chemists, which could be the fixed arms, the mobile manipulator or even an automation platform, its sensory elements and surrounding objects, including laboratory instruments, glassware and particles. The world description is inherently modular and can be set through standardized scripts or USD using Nvidia Isaac Sim's graphical interface.

This section delineates the methods of the four main components of MATTERIX, (1) digital twin environment generation and initialization, (2) multiscale simulation with the integrated physics and semantics engine, (3) workflow plan generation with manipulation skill library, perception and hierarchical state machine and (4) deployment of the simulation to the real setup.

Digital twin environment generation

One of the main bottlenecks in the usage of digital twins for chemistry laboratory automation is the effort required in the generation of those virtual environments and the skill gap between the chemistry domain and digital twin development requirements. To lower this barrier, MATTERIX comes with different methods, including: (1) providing methods to generate digital twin assets and a rich library of assets with commonly used objects and tools in chemistry laboratories and (2) methods to generate digital twin environments with ease.

Asset generation. MATTERIX comes with a library of assets, ready to be used for laboratory automation purposes and to learn new policies in simulation. The following asset types are supported: glassware, laboratory tools and devices, fixed-based and mobile robots, automation platforms and laboratory spaces. Besides rigid and articulated assets, it supports liquid and powders as particle systems, and soft objects.

Digital twin assets are represented using USD[36], a high-performance platform initially developed for film and visual effects. USD supports robust data interchange via comprehensive schemas for geometry, shading, lighting and physics, and its unique composition features enable flexible asset assembly and collaborative workflows. Assets in MATTERIX incorporate physical and textual properties, and users can attach custom attributes-such as object key points and coordinate frames-to enhance simulation and automation tasks.

Physical assets in our digital twin simulation include prims (collision-only static objects), rigid bodies, articulated bodies, robots, particle systems (fluids and powders) and soft objects. In Isaac Sim, powders and fluids are modeled using a position-based dynamics solver[37] integrated with an iterative density solver that enforces constant density via positional constraints. This approach enables large time steps suitable for real-time applications. Consequently, position-based dynamics provides a flexible platform for simulating phenomena such as liquids, granular materials, cloth and deformable bodies.

MATTERIX leverages multiple methods to generate the asset library. One approach is to convert 3D asset models-either user-designed or sourced online-into USD assets, with tools such as Isaac Sim and Blender available for further modification. Robots and articulated models, often described using unified robot description format, can be directly converted to USD in Isaac Sim. In addition, MATTERIX utilizes advances in computer graphics for 3D scene reconstruction. We employ NeRF[33] to generate 3D representations of rigid assets by optimizing a continuous volumetric scene function from sparse input views (images or video scans). Nerfstudio[51] simplifies this process for users with its modular framework and intuitive application programming interface. Online platforms such as Luma AI[52] also offer tools for creating 3D assets. Users manually define core physical properties, including mass. For additional parameters, such as friction coefficient, default values and materials can be refined for any asset-whether NeRF-generated or sourced from 3D assets.

Environment generation. A chemistry digital twin may consist of numerous assets, making environment generation challenging. To simplify this, MATTERIX provides an asset library where users can select reusable models for common laboratory items such as beakers, heaters and robotic arms. As shown in Figs. 1 and 3, users can add assets by selecting them from the library and specifying their location in the environment. The USD scene of an asset in MATTERIX may include multiple objects, such as a rack with pipette tips, reducing duplication and simplifying the creation of complex setups such as the full laboratory environment in Fig. 3. Users can also update additional details such as key points or physical properties, ensuring accurate simulation behavior.

Digital twin environments are created for two main purposes: synthetic environment generation and real-to-sim replication.

The synthetic environment is useful for the feasibility analysis of a workstation before building it in the real world or for developing and testing new methods, such as experimental planning algorithms. A real-to-sim environment replicates a real laboratory setup, supporting tasks such as monitoring and training machine learning models for automation. MATTERIX provides interfaces to connect digital twin environments with real laboratory sensor data, such as camera input and to control robots, as explained later in this section. In addition, MATTERIX supports heterogeneous laboratory setups inherently by enabling the definition of multiple robotic systems (agents), including robots, automation platforms and other devices, allowing concurrent task execution in simulated environments. This multiagent capability facilitates research in scheduling and experimental planning for laboratory automation, paving the way for high-throughput solutions to chemistry automation tasks.

Integrated physics and semantics engine

Many real-world physical and chemical dynamics are challenging to simulate, ranging from the locking mechanism of a door and liquid handling tasks to processes often unsupported by robotics simulators, such as heat transfer and chemical reactions. Despite the ubiquity of such tasks in daily life and chemistry laboratory activities, existing robotic simulation tools do not support them. Furthermore, some assets in the digital twin possess specific functionalities. For instance, a heater, when ?heaterOn ← true acts as a heat source. By integrating these device functionalities, digital twins can effectively manage long-horizon experimental plans. The semantics engine extends the Isaac Sim physics engine by enabling the simulation of these behaviors using approximated models. These behaviors can influence the physics engine's simulation state, and conversely, the physics state can impact the semantics engine. For instance, the amount of water flowing from a faucet in a chemistry laboratory washing station is linked to the degree to which the faucet handle is rotated (Fig. 3f).

The semantics engine can also serve as a source of sparse rewards for training reinforcement learning policies, aiding in the exploration of long-horizon tasks. Implemented in PyTorch with GPU parallelization, the semantics engine enables efficient execution of reinforcement learning algorithms. It is designed to be modular, allowing users to add custom behaviors. MATTERIX offers a semantic behavior library for common chemistry workflows (Fig. 4), including liquid handling, heat transfer and chemical reaction approximations.

The semantics engine is modeled with the logical state vector $\mathbf{l}(t) \in \{0,1\}^m$ and continuous states $\mathbf{s}(t) \in \mathbb{R}^p$, which extends the physics engine's state vector $\mathbf{x}(t) \in \mathbb{R}^n$. The input action vector $\mathbf{a}(t)$ to the MATTERIX simulation consists of both continuous and logical elements, such as robot joint angles for picking up an object or turning on a heater and setting it to a target temperature. Events $\mathbf{l}(t+1) = \mathbf{E}(\mathbf{x}, \mathbf{s}, \mathbf{l}, \mathbf{a})$ and processes $\mathbf{s}(t+1) = \mathbf{P}(\mathbf{x}, \mathbf{s}, \mathbf{l}, \mathbf{a})$ are linked to changes in both instantaneous and continuous semantic states, thereby extending the physics simulation $\mathbf{x}(t+1) = \mathbf{F}(\mathbf{x}, \mathbf{a})$. The logical states $\mathbf{l}(t)$ can influence the behavior of physics dynamics $\mathbf{F}$ and semantic processes $\mathbf{P}$, evolving over time as the logical states change: $\mathbf{F}_t$ and $\mathbf{P}_t$. This can be achieved by either directly updating the models $\mathbf{F}$ and $\mathbf{P}$ or setting preconditions for $\mathbf{P}$ to take effect. For example, loading a pipette tip into the liquid handler requires real-time changes to $\mathbf{F}$, so that when the pipette tip is loaded, it moves in sync with the liquid handler's pipette motion. Similarly, aspirating liquid into the pipette tip requires the tip to be inside the solution. Increasing a heater's temperature, on the other hand, requires the heater to be turned on.

Supplementary Algorithm 1 in Supplementary Section 4 outlines the integration of the physics and semantics engines. While this formalization facilitates the scaling of new behaviors by the semantics engine, MATTERIX also allows users to customize interactions between the physics and semantics engines.

Given that the simulation of heat transfer and chemical kinetics is crucial for simulating complete chemistry workflows, their descriptions are provided in the following paragraphs.

Heat transfer simulation. MATTERIX employs an energy balance approach to predict temperature variations in different components over time, considering conduction and convection as the primary heat transfer modes. Thermal interactions with the surrounding air are modeled through convection, whereas heat transfer between contacting objects is handled via conduction. The conductive heat transfer between two contacting objects ($\dot{Q}_{\mathrm{cond}}$) and the heat transfer through convection ($\dot{Q}_{\mathrm{conv}}$) are modelled by

$$\dot{Q}_{\mathrm{cond}} = \frac{kA}{d}\Delta T \quad \text{and} \quad \dot{Q}_{\mathrm{conv}} = hA\Delta T, \tag{1}$$

where $T(K)$ represents the temperature, k (W m^{-1} K^{-1}) is the thermal conductivity, A (m^2) denotes the heat transfer area, d (m) is the thickness of the conductive medium, h (W m^{-2} K^{-1}) is the convective heat transfer coefficient, $\dot{Q}$ (W) represents the heat transfer rate. Moreover, heat generated by a heat source ($\dot{Q}_{\mathrm{gen}}$) is modeled by

$$\dot{Q}_{\mathrm{gen}} = K_{\mathrm{gen}}(T_{\mathrm{target}} - T(t)), \tag{2}$$

where K_{gen} (W K^{-1}) is the heat source constant, and T_{target} (K) is the target temperature of the heater. By applying the first law of thermodynamics, we derive the governing equations that describe the rate of temperature change for different objects in the digital twin as follows:

$$mC\frac{\mathrm{d}T}{\mathrm{d}t} = \Sigma\dot{Q}. \tag{3}$$

During the simulation, the heat transfer rate values on the right-hand side of equation (3) are determined by the semantic logical states, such as the contact between different objects and whether the heater is turned on. Further details on the heat transfer simulation, along with assumptions and an example, can be found in Supplementary Section 3.

Chemical kinetics simulation. MATTERIX provides approximate reaction kinetics simulations to estimate the change in concentrations of reactants and products as a function of time. This not only verifies the occurrence of a reaction but also permits multistep reactions, in which the product of an earlier reaction is the reactant for a subsequent reaction. For a given reaction, the user specifies the reaction template to transform the reactants in a container into product compounds and also approximate parameters for the kinetics simulation, including the reaction order and the parameters governing the temperature-dependent reaction rate constant, k.

To incorporate temperature dependence, k is calculated at each time step based on the system temperature (T) using the Arrhenious equation[40]

$$k = A\exp\left(-\frac{E_{\mathrm{a}}}{RT}\right). \tag{4}$$

Here, A represents the pre-exponential factor, E_{a} is the activation energy for the reaction, and R is the universal gas constant. It should be noted that the parameters A and E_{a} are estimated based on prior experimental knowledge from organic chemists and should not be taken as exact quantities. To maintain numerical stability, the temperature is constrained to a physically relevant temperature range.

For an example reaction A + 2B → AB_2, the reaction rate would be first order in the concentration of A, denoted [A] and second order in [B]. The reaction rate at a given time would be

$$[\dot{\mathrm{A}}] = \frac{1}{2}[\dot{\mathrm{B}}] = -k[\mathrm{A}][\mathrm{B}]^2, \tag{5}$$

whereas the product concentration reaction rate would be the negative of the reactant concentration reaction rate,

$$[\dot{AB}_2] = +k[A][B]^2. \quad (6)$$

The reaction would continue until the preconditions are no longer met, such as a change in temperature or the depletion of a reactant species.

Although the inclusion of temperature effects enhances the simulation's physical fidelity, several limitations remain, which can be addressed in future improvements. The reactions that can be simulated are limited to forward-only reactions, which can be represented using SMARTS templating. We reiterate that the reaction verification methods are only semiquantitative, and more sophisticated system-specific reaction kinetics simulation and quantum chemical calculations are required for high-quality quantitative predictions.

Reduction–oxidation execution protocol. The following stock solutions were prepared manually using an analytical balance, all chemical were purchased from Sigma-Aldrich.

Solution 0: KI (0.66 g, 4 mmol) and saturated starch in 100 g H_2O, total concentration of KI is 0.04 mol kg^{-1}

Solution 1: H_2O_2 (3 g of 5% w/w, 4.4 mmol) in 100 g H_2O, total concentration of H_2O_2 is 0.044 mol kg^{-1}

Solution 2: ascorbic acid (0.88 g, 5 mmol) in 100 g H_2O, total concentration of ascorbic acid is 0.05 mmol kg^{-1}

Step 1: to a large reaction beaker (beaker 0) containing 100 g of solution 0 (4 mmol I^-, saturated starch) over an analytical balance, the robotic arm picked up beaker 1 containing solution 1 and poured around 50 g of solution 1 into the reaction beaker. An increase in total mass was measured to be 48 g, and the total amount of H_2O_2 added was calculated to be 2.1 mmol. The mixture was left for reaction for 50 s, and the color gradually turned dark blue due to the oxidation of I^- into I_2, which immediately formed a dark blue–brown iodine–starch complex.

Step 2: to the reaction beaker containing product of step 1, the robotic arm picked up beaker 1 containing solution 2 and poured around 50 g of solution 2 into beaker 0. An increase in total mass was measured to be 48 g, and the total amount of ascorbic acid added was calculated to be 2.4 mmol.

The mixture was left for reaction for 50 s, and the dark blue–brown color gradually faded due to the reduction of I_2 into I^-, which lead to the subsequent consumption of the iodine–starch complex.

System and workflow verification. Following the semantics engine description and the workflow logic definition of a chemical workflow, we demonstrate a chemical reaction verification protocol that ensures the simulated execution aligns with expected chemical outcomes. This protocol serves as a verification for both the simulated system and the generated workflow. This protocol tracks the states of compounds within containers, maintaining both identity and concentration throughout the simulated robotic manipulations and reactions. Before executing a user-defined reactions, the preconditions are checked, such as the temperature or presence of reactants. Upon successful precondition checks, reactant compounds are manipulated using cheminformatics templating to produce product molecular compounds. The simulator applies approximate estimations of concentration changes resulting from the reaction, updating the simulated container contents accordingly. It is important to emphasize that these concentration estimations are not intended for quantitative accuracy, which is beyond the scope of this workflow verification. Instead, they provide a qualitative confirmation that the expected reaction has occurred within the defined workflow, aiding in debugging and validation.

This chemical reaction workflow verification protocol serves a dual purpose in the context of automated laboratory procedures. First, it provides a valuable training tool for autonomous agents tasked with executing chemical workflows. By employing a task-based verification method, agents can receive feedback on the correctness of their actions within the simulated environment. This feedback loop is crucial for developing robust and reliable automated agents capable of navigating complex laboratory tasks. Second, the protocol offers chemists a tool for evaluating and debugging in silico representations of their workflows. Even without precise reaction kinetics, the approximate concentration changes and precondition checks provide valuable insights into the feasibility and logical flow of the designed workflow. This allows chemists to identify potential errors or inefficiencies in their experimental plans before physically implementing them in the laboratory.

Workflow plan generation

MATTERIX generates workflow plans by integrating various manipulation and perception methods within a hierarchical task definition framework to construct a skill library. This library enables the easy creation of hierarchical workflows for executing chemistry experiments.

Manipulation methods and skill library. For autonomous execution of chemistry experiments, robotic systems and automation platforms, such as liquid handlers, must be capable of perceiving their environment and manipulating objects and devices. To facilitate this, we developed a skill library that allows users or AI agents to leverage predefined skills to perform long-horizon experiments with multiple sequential steps. The MATTERIX skill library comprises both classical motion planning methods and learning-based policies, all of which are parallelizable on GPUs. These components are described as follows.

For differential inverse kinematics and whole-body mobile robot controller, we employ the damped least-squares method to solve the inverse kinematics problem[53]. This approach utilizes the Jacobian mapping from joint-space velocities to end-effector velocities to compute incremental changes in the joint space, guiding the robot's end effector toward a desired pose. The mobile robot is modeled as a floating base, with different controllers used to move either its base, only the upper body (arm manipulator) or the entire body. In the case of whole-body control, the Jacobians of the robot's end effector are computed relative to the inertial or world frame[54].

For the cuRobo motion planner, to ensure safe robot motion in chemistry laboratories, we employ cuRobo, a collision-free motion generation framework that computes optimized robot trajectories based on the initial configuration and target end-effector pose. The system operates in two key stages: first, solving a parallelized collision-free inverse kinematics problem to determine feasible goal configurations, and second, leveraging GPU-accelerated optimization with parallelized seed trajectories to generate motion solutions. The trajectory optimization formulation incorporates multiple constraints, including joint position, velocity, acceleration and jerk limits, as well as self-collision and environmental collision avoidance, while ensuring the target end-effector pose is reached at the final timestep.

For model-free reinforcement learning, for policy training, we employ proximal policy optimization[55], a policy gradient reinforcement learning method utilizing first-order optimization. Proximal policy optimization maintains an effective balance between data efficiency, reliable convergence properties and implementation simplicity, making it well-suited for robotic applications requiring computational efficiency and robust performance. To train long-horizon policies, MATTERIX employs a curriculum scheduling strategy[56] that begins with simplified short-horizon tasks and progressively increases both task complexity and temporal scope.

Perception. For object pose estimation in chemistry laboratories, we employ FoundationPose[39], a vision foundation model unifying model-based and model-free perception that requires no fine-tuning

for deployment. As our digital twin simulation environments already possess computer-aided design models of all target objects, the system integrates seamlessly. FoundationPose first samples initial poses uniformly around the object, refines them via a neural network and selects the optimal pose with the highest score.

Hierarchical state machine task representation. Automating a chemistry experiment requires collaboration between chemists and automation researchers. Chemists define experimental workflows, while roboticists provide task descriptions and policies that enable structured execution. A chemistry experiment may span multiple workstations or laboratories, and tasks often consist of hierarchically organized subtasks. To accommodate this complexity, MATTERIX represents workflows using a hierarchical state machine, allowing arbitrary levels of abstraction.

A workflow is modeled as a tree $T = \langle N, E \rangle$, where N represents nodes (actions), and E denotes edges connecting them. The tree depth d corresponds to the level of abstraction, with parent nodes representing higher-level tasks and child nodes refining them into specific actions. The leaves of the tree are primitive actions, where execution methods and robotic manipulation policies are implemented. A workflow is completed when all nodes are executed and its goal is achieved. A node is considered executed once all its child nodes have been processed. This structured approach enables seamless integration of policies at different levels of abstraction. Figure 4 illustrates the construction of a reusable skill by integrating manipulation and perception methods within a hierarchical tree structure for task definition. These skills can, in turn, be utilized to create long-horizon workflows.

Deployment to real setup

Upon successful task execution in the simulated digital twin of the chemistry environment, the solution is deployed to the real setup. During deployment, MATTERIX updates the simulated environment with the real setup by interfacing the physical robots and devices, whose states are initially updated in the digital twin to match the real setup. The robot, equipped with an end-effector-mounted camera (or alternatively, a scene camera), observes the laboratory environment and estimates object poses using FoundationPose, which is then used to update the digital twin. Following initialization, the workflow generates actions for the robot and devices in the digital twin, with updated states passed to their physical counterparts for execution after each simulation step.

We employ the real robot in impedance control mode to compensate for minor discrepancies between simulation and reality, particularly during contact-intensive tasks such as pick-and-place or insertion operations. MATTERIX incorporates force feedback from the robot as a safety mechanism, halting task execution if predefined safety thresholds are exceeded. The synchronized digital twin provides a powerful tool for real-time debugging, visualization and monitoring, enabling chemists to track the experiment's progress and facilitating development. Once experimental workflows are validated in simulation, an alternative deployment strategy involves directly executing the verified workflows on the physical systems (robots and devices) without relying on the digital twin.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All data required to evaluate the presented conclusions are available within the Article and its Supplementary Information. All supporting data were generated using the MATTERIX code. Source data are provided with this paper.

Code availability

The code for MATTERIX is available via Zenodo at https://doi.org/10.5281/zenodo.17095671 (refs. 57–63) and via GitHub at https://github.com/AccelerationConsortium/Matterix.

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Author contributions

Authors are listed alphabetically by contribution. Designed and built the core infrastructure: A.M., A.S., G.P., H.F., K.D. and M.B. Rigid assets library: A.M., A.S., B.Z., H.F., J. Chae, J. Choi, K.D., M.B., N.R. and Z.Z. Articulated asset library: A.M., A.S., J. Choi and K.D. Robot assets: A.M., A.S., K.D. and Z.Z. Full laboratory assets: A.M., A.S., J. Choi and K.D. Nested rigid assets: J. Choi and K.D. Environments: A.M., A.S., H.F., J. Chae, K.D., N.R. and Z.Z. Semantics engine: chemistry (A.S., G.T., H.H. and K.D.), device functionalities (A.S. and K.D.) and heat transfer (A.S., H.D. and K.D.). Particle system (fluids and powders): A.S., K.D., M.B. and N.R. Skill library: foundation pose (A.M., S.H. and Y.Z.), inverse kinematics solver (K.D. and M.B.), reinforcement learning (A.M., J. Chae, K.D. and M.B.), whole-body controller (K.D. and Z.Z.) and cuRobo (K.D. and Y.W.). Deployment to real setups: FR3 and OT-2 liquid handling experiment (A.W., H.F., N.R. and S.V.), liquid pouring experiment (A.M. and Y.Z.), pick-and-place experiment (A.M.) and sim-to-real deployment pipeline (A.M., H.F. and K.D.). Writing the paper draft: G.P., H.D., H.F., K.D., M.B., Y.Z. and Z.Z. Supervision: A.A.G., A.G., A.I.C., F.S., G.P., H.F., K.D. and M.B.

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